

Competitive Exam Practice 6: Hardwired Control Unit Design

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CS 1987 Q1-vi, verified — GATEOverflow V3 p.50*] [**MCQ, 1 mark**] A microprogrammed control unit:

- (A) is faster than a hard-wired control unit
- (B) facilitates easy implementation of new instructions
- (C) is useful when very small programs are to be run
- (D) usually refers to the control unit of a microprocessor

Q2 [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a hardwired control unit, every control signal is generated as a function of:

- (A) Opcode decoder lines only
- (B) Step decoder lines only
- (C) Both the opcode lines and the step lines together
- (D) Neither — signals are read from a stored table

Q3 [*Original, GATE-pattern*] [**NAT, 1 mark**] An instruction set architecture has 25 distinct instructions. The minimum number of bits required in the instruction's opcode field to encode all of them is _____.

Q4 [*Original, GATE-pattern*] [**MCQ, 1 mark**] A sequence counter (SC) together with a step decoder, as used to generate one-hot control-step signals in a hardwired control unit, is functionally most similar to:

- (A) A program counter driving an instruction cache
- (B) An instruction register holding the current opcode
- (C) A modulo- n counter feeding an n -output decoder
- (D) A stack pointer

Q5 [*Original, GATE-pattern*] [**NAT, 1 mark**] A hardwired control unit's sequence counter is implemented with 6 bits. The maximum number of distinct control steps it can address is _____.

Q6 [*Original, GATE-pattern*] [MCQ, 2 marks] Which of the following is **NOT** a disadvantage of hardwired control relative to microprogrammed control?

- (A) Difficult to modify after the chip is fabricated
- (B) Design/verification complexity grows with instruction-set size
- (C) Slower per control step than microprogrammed control
- (D) Requires re-deriving Boolean equations whenever the ISA changes

Q7 [*Original, GATE-pattern*] [MCQ, 2 marks] For a control signal shared by two instructions at the same control step, e.g. $Z_{\text{in}} = (D_{\text{ADD}} \cdot T_2) + (D_{\text{SUB}} \cdot T_2)$, the OR operation reflects the fact that:

- (A) ADD and SUB must execute simultaneously
- (B) Z_{in} must be asserted whenever *any* instruction whose state table lists it at that step is running
- (C) Z is a shared bus line, not a register
- (D) The equation is redundant and can be simplified to a single term